

NICE on familial hypercholesterolaemia

1. DIAGNOSIS

NICE 2008 (updated 2019) CG71



Red Whale

GEMS

Guidelines & Evidence Made Simple

NICE asks us to identify new potential cases when filing cholesterol results/taking a family history, and to systematically identify patients at high risk by searching our records for those with high cholesterol.

Identify new potential cases based on cholesterol levels/personal or family history

Systematic search for those at high risk

From history
CHD under 60y
in patient
or
first-degree
relative

From cholesterol levels
(>16y)
Total cholesterol >7.5

Total cholesterol >9 if ≥30y
or
>7.5 if <30y

Why so high? NICE balanced the likelihood of having FHc against the time/cost effort involved. These levels are >99th centile for age, so highly likely to have FHc. Lower thresholds would result in more false positives and significant additional workload/costs

Suspect HOMOZYGOUS FHc if:

LDL >13 if ≥16y

LDL >11 if <16y

AND REFER

(do not use Simon Broome/Dutch criteria)

Assess likelihood of FHc: use Simon Broome criteria
or
Dutch Lipid Clinical Network criteria

REFER for genetic testing, management and 'cascade testing' of relatives if:

Simon Broome or Dutch Lipid Clinical Network criteria met OR HOMOzygous disease suspected

Simon Broome criteria

Total cholesterol >7.5 or LDL >4.9 (If <16y: total cholesterol >6.7 or LDL >4)	PLUS any ONE of	Tendon xanthomas (rarely appear <30y) in patient or 1 st or 2 nd degree relative
		Family history of MI <60y in 1 st degree relative <50y in 2 nd degree relative
		High cholesterol in 1st or 2nd degree relative Total cholesterol >7.5 in adults Total cholesterol >6.7 in children
OR		
Confirmed mutation		

Dutch Lipid Clinical Network criteria (easier using online calculator, e.g. MDCalc)

'Premature' is <55y (males) and <60y (females)

		Score
1 st degree relative with...	Premature CHD/CVD or LDL >95 th centile (on chart)	1
	Tendon xanthomas/arcus OR <18y with LDL >95 th centile (on chart)	2
Clinical history	Premature coronary artery disease	2
	Premature cerebral/peripheral vascular disease	1
Physical findings	Tendon xanthoma	6
	Arcus cornealis <45y	4
LDL	LDL ≥8.5	8
	LDL 6.5–8.4	5
	LDL 5.0–6.4	3
	LDL 4.0–4.9	1
DNA	Mutation in relevant genes	8

Those with tendon xanthomas or confirmed mutation have definite FHc. The others have possible FHc but still need referral

REFER if Simon Broome criteria met or Dutch Lipid score >5

Which criteria to use? NICE says either can be used. Simon Broome is easier but is based on a UK database started in the 1980s; the BMJ suggested it may not be as applicable to other populations, including Asian populations, who tend to have lower cholesterol (BMJ 2023;382:e073280).

**Referral**

- Referral should be offered to all with suspected FHc. This is for:
 - Initiation of treatment.
 - Assessment of personal and family risk.
 - Cascade testing to identify asymptomatic relatives.
- If there are symptoms/signs of coronary heart disease, refer to cardiologist.
- In those with suspected HOMOZYGOUS FHc:
 - Refer to specialist centre: LDL apheresis/liver transplant may be offered.

NICE doesn't recommend testing for secondary causes, but the BMJ review suggests:

- Hypothyroidism.
- Liver disease.
- Renal disease.
- Ketogenic diet.
- Medication, e.g. highly-active retroviral drugs.

Cascade testing

This involves using the index case to identify other potentially-affected family members through detailed family history and testing. It is a systematic secondary care process: a BMJ review suggested that '*Tell your relatives to get tested*' was not effective! (BMJ 2012;344:e3228) **NICE recommends:**

- In families where a mutation has been identified**, genetic testing should be offered to those at risk (don't use LDL concentration in these individuals). If found NOT to carry the gene, manage CV risk as you would for the rest of the population (80% will have a known mutation).
- In families where a mutation has NOT been identified**, use age- and sex-standardised LDL concentrations to diagnose family members with FHc (provided in the NICE guidance but not here because this is secondary care territory). Do not use the Simon Broome criteria in these people as it underestimates risk.
- Consider screening all first- and second-degree relatives. Include third-degree relatives if possible** (great grandparents, great grandchildren, great aunts/uncles/niece/nephews, first cousins).

Testing children of affected adults

If one parent is affected: test offspring by age 10y (or as soon as possible after this):

- Offer DNA testing if family mutation present.
- Measure LDL concentration if no known family mutation. Repeat after puberty to exclude diagnosis.

If both parents are affected (or clinical signs in child): test by age 5y (or as soon as possible after this) by measuring LDL concentration. If LDL >11, consider diagnosis of FHc and manage appropriately.

Management

- Aim is to reduce LDL by more than 50% from pretreatment levels.**
- All the usual lifestyle advice should be offered (smoking, diet, activity, weight management, alcohol).
- Never use a risk estimator (e.g. QRISK3) to assess risk: these tools will significantly underestimate risk!
- Statins are first line in adults and children. Ezetimibe may be considered alongside statins in those with FHc (or as monotherapy if statins not tolerated/contraindicated) (NICE 2016, TA385). Treatment is for life.
- PCSK-9 inhibitors may be initiated by specialist if NICE criteria met (TA393, TA394 (2016), TA733 (2021)).

	Evolocumab or alirocumab	Inclisiran
In those WITHOUT CVD	LDL persistently >5mmol/l	Research trials only
In those WITH CVD	LDL persistently >3.5mmol/l	LDL persistently ≥2.6mmol/l

More on these drugs in the Lipids and statins article.

Females: NICE says COCP not contraindicated, but it may increase CV risk so alternatives may be preferred. Statins are not recommended during pregnancy (risk of congenital malformation) or breastfeeding (but women can breastfeed on bile acid sequestrants).

In children with FHc: specialist will offer statins by the age of 10y or at the earliest opportunity thereafter.

Annual review

- Check smoking status (NICE doesn't mention other CV risks, e.g. BP/lifestyle, but logical to include!).
- Check LDL levels, and review medication side-effects and compliance.
- Review progress of cascade testing and update family history (new CHD events, etc.).
- Look for symptoms/signs of CHD (consider an ECG).